A Vector Derivation of Miller Indices

The process to determine Miller Indices is first to find the intercepts of the plane with the crystal axes then to take the reciprocals of these intercepts. The plane can then be labeled by the reduced integers found. The purpose of this document is to provide a derivation of this process from vector math. Beginning with the process described above in the most general case we start with the intercepts $(x, 0, 0)$, $(0, y, 0)$, and $(0, 0, z)$. Taking the reciprocal of each non-zero component and associating the value with the appropriate direction yields the plane and normal vector

$$\begin{matrix} & (\frac{1}{x} & \frac{1}{y} & \frac{1}{z}) \\ \vec{n} = [\frac{1}{x} & \frac{1}{y} & \frac{1}{z}] \end{matrix}$$

These values are reduced to the final result by multiplying by the greatest common factor of x, y, and z. Let's call this factor as m_{GCF}. Note that $\hat{n}$ is used here to mean the vector is reduced to its smallest integer value rather than having a magnitude of 1.

$$\begin{matrix} & m_{GCF}(\frac{1}{x} & \frac{1}{y} & \frac{1}{z}) \\ \hat{n} = m_{GCF}[\frac{1}{x} & \frac{1}{y} & \frac{1}{z}] \end{matrix}$$

The process of quickly going from the intercepts to the plane works can be demonstrated simply using vectors. A plane can be determined by any three points in the plane that are not collinear by taking the cross product of two vectors pointing between the points. Since we are interested in points that are intercepts they are not collinear and we can again use the points $(x, 0, 0)$, $(0, y, 0)$, and $(0, 0, z)$. Without loss of generality we can create two vectors pointing to the x-intercept from the other points,

$$\begin{aligned} \vec{v_1} &= [x \quad -y \quad 0] \\ \vec{v_2} &= [x \quad 0 \quad -z] \end{aligned}$$

Taking the cross product $\vec{v_1} \times \vec{v_2}$ we get a vector normal to the plane

$$\vec{n} = [yz \quad xz \quad xy]$$

Factoring out xyz the normal vector can be put in a familiar form.

$$\vec{n} = xyz[\frac{1}{x} \quad \frac{1}{y} \quad \frac{1}{z}]$$

Now, we can determine $\hat{n}$ as before.

$$\hat{n} = m_{GCF}[\frac{1}{x} \quad \frac{1}{y} \quad \frac{1}{z}]$$

This demonstrates where the reciprocal procedure used to determine Miller Indices comes from. Miller indices and the math underlying them are crucial for describing and analyzing crystallographic structures and properties.